Name:

Master EMARO

Mobile Robots Exam

2 h - Allowed documents: mementos only. No laptop, no phone, no calculator.

Exercise 1 (10 mins): Parametrization. **Answer in the provided table.**

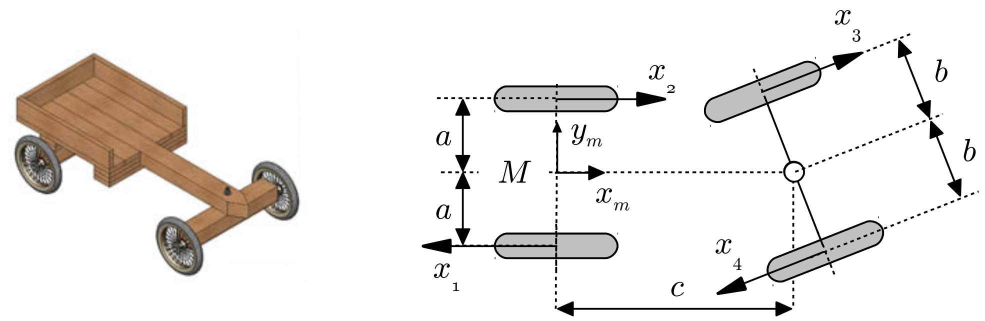


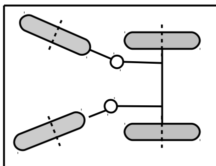
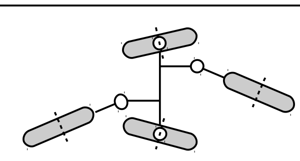
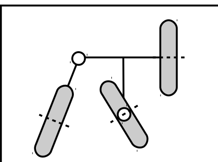
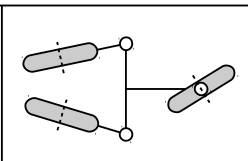
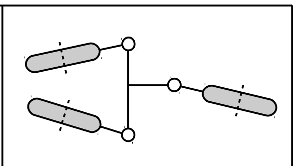
Figure 1. A « soapbox » and its schematic representation.

Wheels 3 et 4 can be represented as two general wheels. In the parameter table, the fact that the wheels are rigidly linked will be reflected by replacing β_4 by its expression as a function of β_3.
Fill the parameter table of the robot :

Wheel	L	α	d	β	γ	φ	ψ=α+β+γ

Exercise 2 (10 mins): Kinematics. **Answer on the subject sheet.**

Correct answer: 100 %, no answer: 0 %, incorrect answer: -50 %. No justification required.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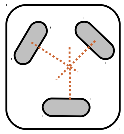
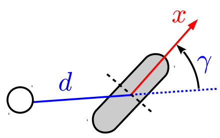
Circle/tick the appropriate answer or answer the question in the rightmost column. All questions under pure rolling constraints. Correct answer: 100 %, no answer: 0 %, incorrect answer: -50 %.		
1. As castor wheels do not generate kinematic constraints, it is possible to replace any wheel by a castor wheel without changing the type of the robot.	true	false
2. Depending on the other wheels of the mobile robot and their actuation, a steerable wheel may or may not have an orientation actuator.	true	false
3. Fixed wheels do not necessarily have a spin actuator.	true	false
4. The non slipping constraints (v_t=0) of conventional wheels play no role in the determination of the degree of mobility of a robot.	true	false
5. The non skidding constraints (v_n=0) of castor wheels must be taken into account when determining the robot type.	true	false
6. It is possible to build an omnidirectional robot with steerable wheels only.	true	false
7. What is the degree of mobility of this wheel configuration (no calculation, no justification): 	Answer:	
8. This general wheel ($d \neq 0$, $\gamma \neq 0$) limits the robot mobility just as a steerable wheel. 	true	false
9. Name two types of wheels which can be used to build an omnidirectional robot.	Answers:	
10. A robot has three steerable wheels : a. The number of independant wheels is:…………………………………………… b. The **theoretical minimum** number of wheel actuators (orientation actuators included) is:……………………………………………………		

Exercise 3 (30 mins): Kinematic modelling

We are going to study the mobile robot defined by the following table of parameters:

W	L	α	d	β	γ	φ	ψ
$1f$	a	$-\pi/2$	0	0	$\pi/2$	φ_{1f}	0
$2f$	a	$\pi/2$	0	0	$-\pi/2$	φ_{2f}	0
$3s$	b	0	0	β_{3s}	0	φ_{3s}	β_{3s}

Table 3.1. Parameters of the robot

In this exercise, failure to answer question 3.1. does not prevent from doing the rest of the exercise.

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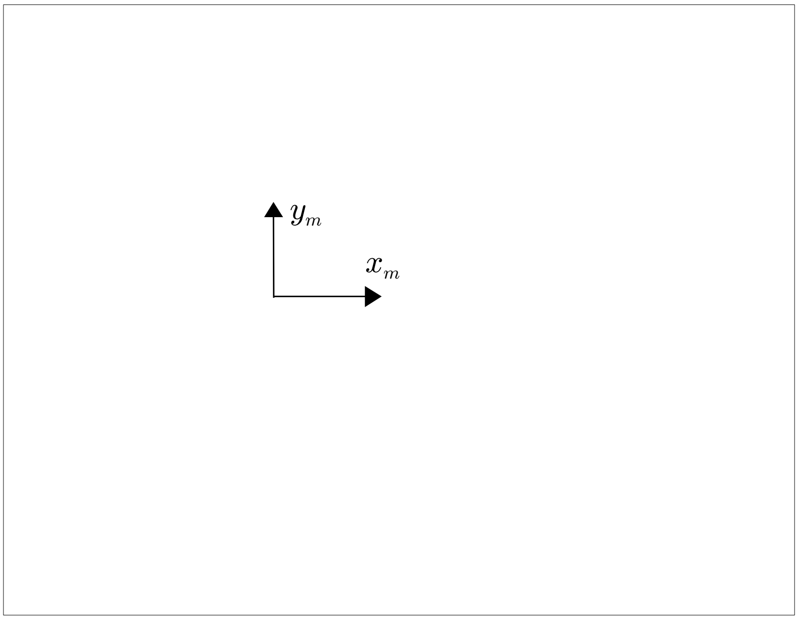
3.1. Give the schematic representation of the robot. The figure should show vectors x_1, x_2 and x_3, the dimensions a and b, and the angle β_{3s}. Make sure β_{3s} is indicated by a circular arrow with the origin at the zero position of the angle. The position and orientation of frame R_m is imposed in the figure and you must comply with it. Fit the schematics in the marked rectangle. I recommend you first draw on scrap paper and copy a **clean result** on the answer sheet once you are sure of your result. Clarity and cleanliness of the figure will be evaluation criteria.



Cleanly draw your robot in this box, complying with the definition of R_m.

The following questions will be solved on the exam sheets provided by the University for the exam. It is your responsibility to make sure that you return both the exam sheets and the subject sheets, which also contain some of your answers.

3.2. Define the configuration vector of the robot.

3.3. Calculate matrices J_1, J_2, C_1 and C_2 of the robot.

3.4. Define matrix C_1^* and **calculate** its rank. The rank of C_1^* must be clearly and rigorously established. A mere statement of the rank is not enough. It is **simple and fast**, but rigorous mathematical proof of the rank is required.

3.5. Give the kernel of C_1^*. Again, it is simple and fast but a mere statement of the kernel is not enough if it is not rigorously established.

Exercise 4 (20 mins): Measurement equations

We study a localization system based on the detection of the transitions between the squares of a checkerboard floor using infrared sensors. In essence, the output of the sensors over a light/dark surface is 1/0. So changes of sensor state correspond to dark-to-bright or bright-to-dark transitions. The robot uses two sensors S_1 and S_2 located at know positions in R_m (see figure 4.1).

The transition lines of the checkerboard floor are the beacons of the localization system. So when establishing a measurement equation, you assume as usual that you know exactly which transition has been detected. The transition is either a line x = know constant or a line y = known constant.

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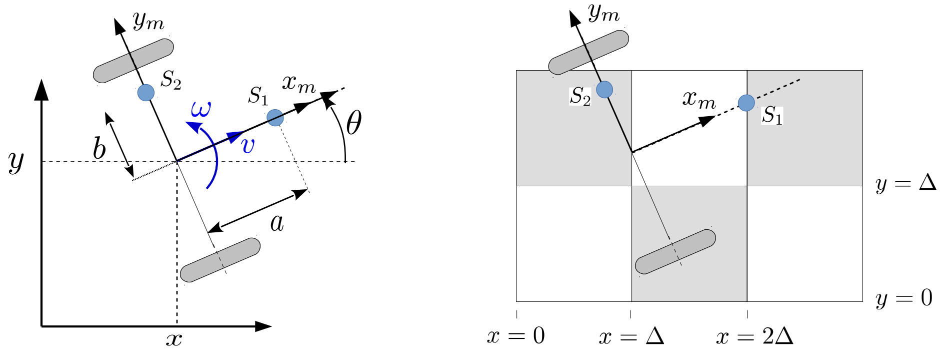

(a) Sensors and notations (b) Sensors and checkerboard surface

Figure 4.1

Answer on the subject sheet. If you need to, use scrap paper to establish the results and then write the final result on the present sheet. If you need intermediate calculations at some point, there is no need to put them here, just put the final result.

3.1. What are the coordinates of the sensors in R_m, ${}^mS_1$ and ${}^mS_2$?

${}^mS_1 =$ ${}^mS_2 =$

3.2. When the robot is at $(x, y, \theta)^t$, what are the coordinates of the sensors in R_o, ${}^oS_1$ and ${}^oS_2$?

${}^oS_1 =$ ${}^oS_2 =$

3.3. When sensor S_1 crosses a vertical line x=$n\Delta$, what is the value of the x coordinate of S_1 in R_o?

x coordinate of S_1 when crossing line x=$n\Delta$ is:

3.**4.** The state vector is $X = (x, y, \theta)^T$. Δ is a known constant.

a. Write the following observation equations:

- g_{1x}: Corresponding to sensor S_1 crossing a “vertical” line (x=$n\Delta$, n know constant).

  $g_{1x} =$

- g_{1y}: Corresponding to sensor S_1 crossing a “horizontal” line (y=$n\Delta$, n know constant).

  $g_{1y} =$

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- g_{2x}: Corresponding to sensor S_2 crossing a “vertical” line (x=$n\Delta$, n know constant).

  $g_{2x} =$

- g_{2y}: Corresponding to sensor S_2 crossing a “horizontal” line (y=$n\Delta$, n know constant).

  $g_{2y} =$

b. Write the C matrices of the Kalman filter corresponding to these measurement equations.

$C_{1x} =$

$C_{1y} =$

$C_{2x} =$

$C_{2y} =$

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Exercise 5 (40 mins): Observability study

This exercise may be a bit more difficult than the previous ones. It is also longer, but its weight (number of grade points) will be the same as exercise 3. It is also less guided. So you are advised to solve the present exercise once you are sure of what you did in all previous exercises, which have a more favorable "grade points to time" ratio. In the grading, special attention will be given to correct definition of the problem, rigorous derivation of results and thoroughness/completeness of the study.

Answer on University exam sheets.

Consider the robot of figure 5.1. The posture of the robot is defined with the usual notations. The velocity of point B, center of the steerable wheel, is in the plane of the wheel and its speed is v.

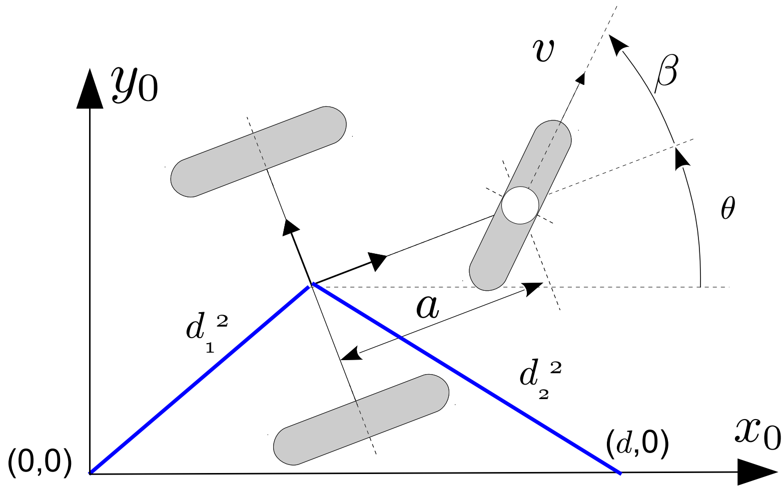

Figure 5.1.

The kinematics of the robot are given by the following set of equations:

$$\begin{cases} \dot{x} = v \, \cos\beta \, \cos\theta \\ \dot{y} = v \, \cos\beta \, \sin\theta \\ \dot{\theta} = (v/a) \, \sin\beta \end{cases}$$

Without loss of generality, we will take $a = 1$ and $d = 1$.

The robot is equipped with two sensors which measure the **squared** distance from point M to the origin of the absolute reference frame and to point (d,0) of the absolute reference frame.

The robot and its sensors are to be used in a localization problem. The aim is to determine the posture of the robot using an EKF. You are required to perform an observability study, in order to anticipate the possible problems that may be encountered.

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Exercise 6 (10 mins): Observability questions

Circle/tick the appropriate answer.
Correct answer: 100 %, no answer: 0 %, incorrect answer: -50 %.

In this exercise, we consider the observability of the following continuous non linear system:

$$\begin{cases} \dot{x} = f(x,u) \ , \ x \in \mathbb{R}^n \\ y = g(x) \ , \ y \in \mathbb{R}^p \end{cases}$$

1. The system cannot be observable if $p < n$.	true	false
2. Generally, observability of the system depends on the input u.	true	false

3. Observability is:
   - a. The ability to reconstruct x and y from u……………………………………□
   - b. The ability to reconstruct x from u…………………………………………□
   - c. The ability to reconstruct x from y and u…………………………………...□
   - d. The ability to detect incorrect measurements in y……………………………□
   - e. The ability to reconstruct missing values of u using x and y…………………□

4. The rank condition is:
   - a. A necessary condition for the system to be observable………………………□
   - b. A sufficient condition for the system to be observable………………………□
   - c. A necessary and sufficient condition for the system to be observable…...…...□
   - d. None of the above, it's a useful indication but nothing more…………………□

5. Assuming the system is observable:
   - a. Any state observer will converge……………………………………………□
   - b. An Extended Kalman Filter (EKF) will not necessarily converge.…………□
   - c. The fact that any particular state observer converges is not guaranteed………□